User Acceptance Testing Test Cases

Beta cinema booking reliability enhancement initiative

05/2026

Document revisions

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28/05/2026	1.0	Initial document

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1 Purpose

This document defines the User Acceptance Test (UAT) cases used to validate that the Beta Cinema booking system meets the business and functional requirements defined in the Business and System Requirements Document (BRSD), version 1.0 (05/2026). The test cases cover the two core problems addressed by the project: the self-locking seat defect and unissued tickets after successful payment.

2 Scope

Test cases in this document trace directly to the Functional Requirements (FR-01 to FR-07), Business Rules (BR-01 to BR-08), and selected Non-Functional Requirements (NFR-01 to NFR-07) defined in the BRSD. Each test case includes preconditions, step-by-step actions, test data, and the expected result, so that any tester or stakeholder can execute it without ambiguity.

3 Test case summary

Test case ID	Title	Related	Priority
TC-01	Immediate seat release when user explicitly cancels payment	FR-01, BR-03	High
TC-02	Seats are preserved when the payment gateway returns a failed transaction	FR-02, BR-04	High
TC-03	User resumes an interrupted booking session after back-navigation	FR-03, BR-02	High
TC-04	User resumes booking after a network disconnection	FR-03	High
TC-05	Seats are auto-released when the holding period expires without resume	FR-03, BR-01	High
TC-06	Booking completes successfully when payment confirmation is received normally	FR-04, BR-05	High
TC-07	System retries payment callback after a transient network drop	FR-04, NFR-01	High
TC-08	User sees a pending status screen when ticket issuance is delayed	FR-05, BR-07	High
TC-09	Ticket is generated within 3 seconds under normal load	NFR-03	Medium

TC-10	User receives booking confirmation via Email/SMS for a delayed transaction	FR-06, BR-08	High
TC-11	Late payment confirmation after the 10-minute window does not trigger automatic ticket issuance	FR-04, BR-06	High
TC-12	Payment/booking anomaly is logged and triggers a real-time alert on database failure	FR-07	Medium
TC-13	Booking token / session ID expires automatically after the holding period	NFR-06	Medium

4 Detailed test cases

4.1 Self-locking seat defect

4.1.1 Test case 1.1

Test Case ID	TC-01
Title	Immediate seat release when user explicitly cancels payment
Related Requirement	FR-01, BR-03
Priority	High
Preconditions	User has selected seat A1, A2 and is on the payment screen within the 10-minute holding window.
Test Steps	1. User selects seats A1, A2 on the seat map. 2. User proceeds to the payment screen. 3. User taps the “Cancel” button on the payment screen. 4. Another user (or the same user via a second session) attempts to select seats A1, A2 immediately after cancellation.
Test Data	Seats: A1, A2. Holding window: 10 minutes.
Expected Result	Seats A1 and A2 are released back to the available pool within 1 second of cancellation (per NFR-05) and can be re-selected immediately by any user, including the original one.
Status	Not Executed

4.1.2 Test case 1.2

Test Case ID	TC-02
Title	Seats are preserved when the payment gateway returns a failed transaction
Related Requirement	FR-02, BR-04
Priority	High
Preconditions	User has selected seats and submitted a payment that will be rejected by the gateway (e.g., insufficient balance).
Test Steps	1. User selects seat B5 and proceeds to payment. 2. User chooses a payment method and submits the transaction. 3. The payment gateway returns a "failed" result. 4. User checks the seat map / payment screen.
Test Data	Seat: B5. Payment result: Failed.
Expected Result	Seat B5 remains held under the user's session for the remainder of the 10-minute holding window, and the user can retry payment without re-selecting the seat.
Status	Not Executed

4.1.3 Test case 1.3

Test Case ID	TC-03
Title	User resumes an interrupted booking session after back-navigation
Related Requirement	FR-03, BR-02
Priority	High
Preconditions	User has selected seat C3 and navigated to the payment screen, then pressed the device/browser back button before the 10-minute window expires.
Test Steps	1. User selects seat C3 and proceeds to the payment screen.

	2. User navigates back to the seat selection screen using back-navigation. 3. User re-opens the same showtime within the holding window.
Test Data	Seat: C3. Elapsed time after back-navigation: 3 minutes.
Expected Result	Seat C3 is still shown as held by the same user (session-bound) and is automatically re-selected/highlighted, allowing the user to continue the booking without re-locking or being blocked.
Status	Not Executed

4.1.4 Test case 1.4

Test Case ID	TC-04
Title	User resumes booking after a network disconnection
Related Requirement	FR-03
Priority	High
Preconditions	User has confirmed seat selection and is on the payment screen; the holding timer is active.
Test Steps	1. User selects seat D7 and confirms within 10 minutes. 2. User's network connection drops for 2 minutes. 3. User reconnects to the internet within the holding window and reopens the app.
Test Data	Seat: D7. Disconnection duration: 2 minutes (within the 10-minute window).
Expected Result	The system recognizes the returning session via the booking token/session ID and resumes the booking at the payment step without releasing seat D7.
Status	Not Executed

4.1.5 Test case 1.5

Test Case ID	TC-05
Title	Seats are auto-released when the holding period expires without resume
Related Requirement	FR-03, BR-01
Priority	TC-05
Preconditions	User has held seat E1 but does not complete payment or return to the app.
Test Steps	1. User selects seat E1 and does not proceed further. 2. Wait for the 10-minute holding period to fully elapse without any user action. 3. Another user attempts to select seat E1.
Test Data	Seat: E1. Idle time: 10 minutes 5 seconds.
Expected Result	Seat E1 is automatically released exactly at the 10-minute mark and becomes available for other users to select.
Status	Not Executed

4.1.6 Test case 1.6

Test Case ID	TC-06
Title	Booking token / session ID expires automatically after the holding period
Related Requirement	NFR-06
Priority	Medium
Preconditions	A booking session and token were created when the user selected a seat.
Test Steps	1. User selects a seat, generating a session ID / booking token. 2. Allow the 10-minute holding window to fully elapse without completing the booking. 3. Attempt to reuse the same session ID / booking token to access the booking.
Test Data	Holding window: 10 minutes.

Expected Result	The session ID / booking token is invalidated automatically once the holding period ends, and any further request using it is rejected.
Status	Not Executed

4.2 Unissued tickets post-payment

4.2.1 Test case 2.1

Test Case ID	TC-07
Title	Booking completes successfully when payment confirmation is received normally
Related Requirement	FR-04, BR-05
Priority	High
Preconditions	User has selected a seat and chosen MoMo as the payment method.
Test Steps	1. User completes payment via the MoMo gateway within the holding window. 2. The gateway sends a successful payment callback to the system. 3. The system validates the payment confirmation.
Test Data	Payment method: MoMo. Result: Success, received within 5 seconds.
Expected Result	The system validates the callback, proceeds to ticket generation, and the booking is marked completed with no data loss.
Status	Not Executed

4.2.2 Test case 2.2

Test Case ID	TC-08
Title	System retries payment callback after a transient network drop
Related Requirement	FR-04, NFR-01
Priority	High

Preconditions	Payment is successfully deducted, but the first callback from the gateway to the system fails to arrive due to a simulated network error.
Test Steps	1. User completes payment via VNPAY. 2. Simulate a dropped connection on the first callback attempt. 3. Observe the system's retry behavior.
Test Data	Payment method: VNPAY. Simulated network failure on attempt 1.
Expected Result	The system automatically retries the callback synchronization at least 5 times (per NFR-01) until it successfully receives and processes the payment confirmation, with no manual intervention required.
Status	Not Executed

4.2.3 Test case 2.3

Test Case ID	TC-09
Title	User sees a pending status screen when ticket issuance is delayed
Related Requirement	FR-05, BR-07
Priority	High
Preconditions	Payment has been confirmed successfully, but ticket generation takes longer than expected due to high system load.
Test Steps	1. User completes payment successfully. 2. Simulate a delay of more than 3 seconds in the ticket generation process. 3. Observe the screen shown to the user.
Test Data	Simulated ticket generation delay: 8 seconds.
Expected Result	Instead of being redirected to the Home screen with no feedback, the user sees a "Ticket Pending" screen stating that the ticket is being generated and will be sent via Email/SMS.
Status	Not Executed

4.2.4 Test case 2.4

Test Case ID	TC-10
Title	Ticket is generated within 3 seconds under normal load
Related Requirement	NFR-03
Priority	Medium
Preconditions	Payment has been confirmed successfully under normal (non-peak) system load.
Test Steps	1. User completes payment successfully. 2. Measure the time from payment confirmation to ticket generation.
Test Data	Load condition: Normal traffic.
Expected Result	The digital ticket is generated and the booking confirmation screen is displayed within 3 seconds of payment confirmation.
Status	Not Executed

4.2.5 Test case 2.5

Test Case ID	TC-11
Title	User receives booking confirmation via Email/SMS for a delayed transaction
Related Requirement	FR-06, BR-08
Priority	High
Preconditions	A booking was placed in a pending state due to delayed ticket issuance (as in TC-08).
Test Steps	1. Continue from the pending state in TC-08. 2. Wait for the system to complete ticket generation in the background. 3. Check the registered email/phone number for a notification.
Test Data	Notification channel: Email and SMS.
Expected Result	Within 15 minutes of successful payment confirmation (per NFR-02), the user receives the final booking status and ticket details via Email or SMS.
Status	Not Executed

4.2.6 Test case 2.6

Test Case ID	TC-12
Title	Late payment confirmation after the 10-minute window does not trigger automatic ticket issuance
Related Requirement	FR-04, BR-06
Priority	High
Preconditions	User initiates payment near the end of the holding window; the gateway confirmation arrives after the window has already expired.
Test Steps	1. User selects a seat and initiates payment at minute 9:50 of the holding window. 2. Simulate the payment gateway returning a success confirmation at minute 10:10 (after expiry). 3. Observe the system's handling of the late confirmation.
Test Data	Payment confirmation arrival: 10 minutes 10 seconds after seat selection.
Expected Result	The system does not automatically issue a ticket. The transaction is flagged and routed to the Reconciliation Queue for manual review or refund, and the customer is notified that the transaction is under review.
Status	Not Executed

4.2.7 Test case 2.7

Test Case ID	TC-13
Title	Payment/booking anomaly is logged and triggers a real-time alert on database failure
Related Requirement	FR-07
Priority	Medium
Preconditions	Payment is confirmed successfully, but a simulated database error (Error 500) occurs while saving the ticket.

Test Steps	1. User completes payment successfully. 2. Simulate a Database Error 500 during ticket/seat status save. 3. Check the system error log and the technical team's alert channel.
Test Data	Simulated error: Database Error 500.
Expected Result	The anomaly is recorded in the error log with full transaction details, and a real-time alert is sent to the technical support team within 60 seconds (per NFR-07).
Status	Not Executed

5 Sign-off

Upon successful execution of all High-priority test cases above with no critical defects, the booking reliability enhancement initiative is considered ready for User Acceptance Testing sign-off and production deployment.